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Model Predictive Control System with Single-Shot Object Detection for Automobile Safety

Autonomous vehicles are driving passengers more than ever, especially with the increase in electric cars and the movement for a cleaner environment. However, a critical issue remains among these vehicles: unreliability due to unpredictable maneuvers from autonomous decision-making systems, leading to a significantly higher rate of road accidents. Due to this spike in driverless cars, autonomous collision avoidance requires better anticipation and pathfinding systems to improve overall safety within these automobiles.

This project aims to design, manufacture, and test a model VEX robot that utilizes Odometry tracking, Dynamic Window Approach path planning, Model Predictive Control path following, and Single-Shot Detection AI object recognition to fabricate a safe environment with predictability and caution. Furthermore, several other algorithms were coded and compared, such as Pure Pursuit path following, to ensure this project's new technology provides safety in the real world. Extensive amounts of data gathered from the robot were analyzed and validated using simulations in QT Creator 15.0.0. Quantitative results from these simulations show that the robot avoided 98% of the presented obstacles with a path deviation of 0.323%, improving upon the 1.936% path deviation concluded from other studies.

Results from testing on various courses show an overall success in its ability to safely maneuver to its desired location, with enough speed and precision to rival current driverless cars. This accomplishment emphasizes the need for improved local predictive control systems to be implemented within autonomous vehicles to improve overall safety in driverless cars.